

A collection of historical artifacts is arranged on a light-colored, textured surface. In the top left, a portion of a wooden chessboard with a blue and brown checkered pattern is visible, featuring several chess pieces. Below the chessboard, there are two ornate medals. The top medal has a red ribbon with a circular red emblem. The bottom medal has a blue ribbon with a circular blue emblem. To the right of the medals is a pair of thin-framed, round-rimmed glasses. In the bottom left corner, a small, round, silver-colored compass is visible, showing a compass rose with cardinal and intercardinal directions. The background is a plain, light-colored surface.

Information and Interface Agents

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Outline

- Introduction
 - ◆ Information discovery and brokerage
 - ◆ Information retrieval
 - ◆ Information integration
 - ◆ Information filtering
 - ◆ Information service and delivery



Information agent tasks

- ◆ Information retrieval
- ◆ Information classification/filtering
- ◆ Information monitoring
- ◆ Information processing
 - summarization/compression/decompression
/translation
- ◆ Information extraction/integration
- ◆ Information management
- ◆ Information brokerage/facilitators
- ◆ Information delivery



Natural language processing and information agents

- ◆ Parsing techniques
- ◆ Corpus techniques
- ◆ Learning language models (statistical techniques)
 - Collocation
 - Mutual information

Information discovery and seeking

- ◆ Internet softbot
- ◆ Internet worms, crawlers



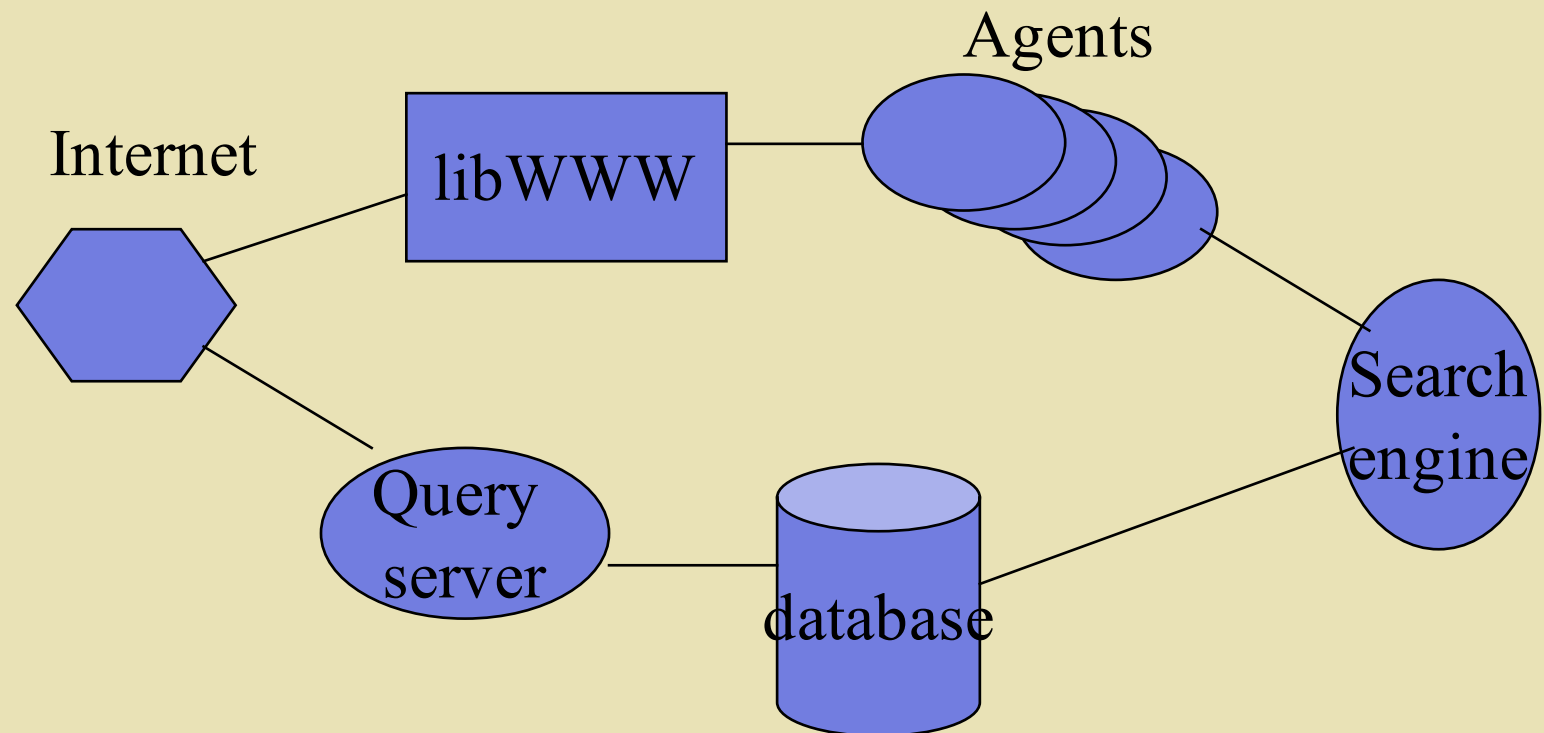


WWW & GENVL— World Wide Web Worm(94)

- ◆ WWW -- A web resource location tool
<http://www.cs.colorado.edu/home/mcbryan/WWW.html>
- ◆ GENVL(Generate Virtual Library)—
interactive hierarchical virtual library for
cataloguing web resources by subject area.
http://www.cs.colorado.edu/homes/mcbryan/public_html/bb/summary.html

Webcrawler[Brian Pinkerston]

- ◆ Index both document titles and contents
use vector space model



Softbot [Etzoni]

- ◆ <http://www.cs.washington.edu/research/projects/softbots/www/presentations.html>

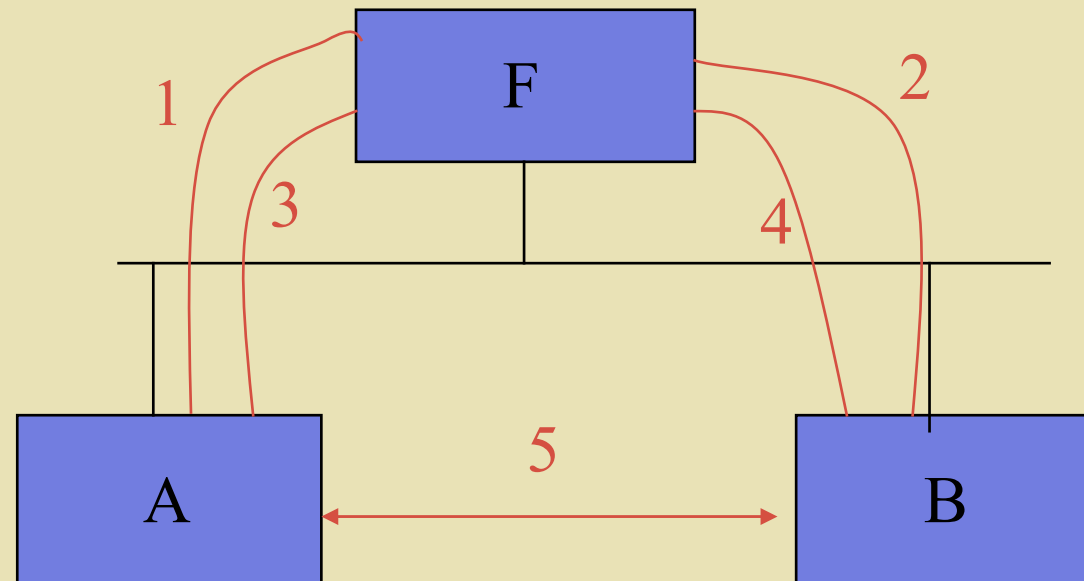




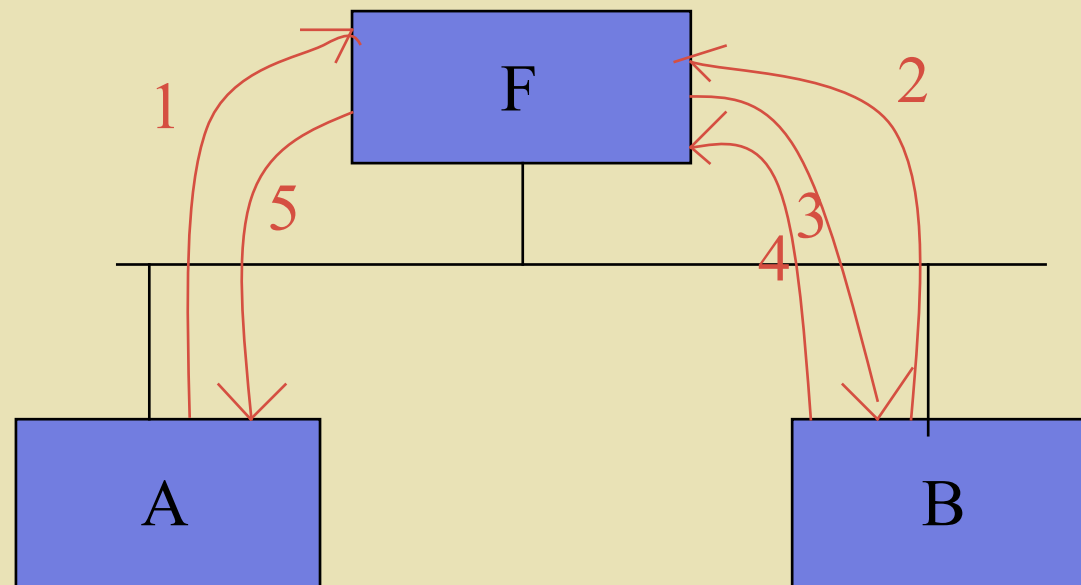
Information facilitator agents

- ◆ Name Server (White Page)
- ◆ Agent Capability Server (Yellow Page)
- ◆ Content-based Routing
- ◆ Translation
- ◆ Problem Decomposition
- ◆ Monitoring
- ◆ Buffering

Matchmaker agent



Broker agents





Outline

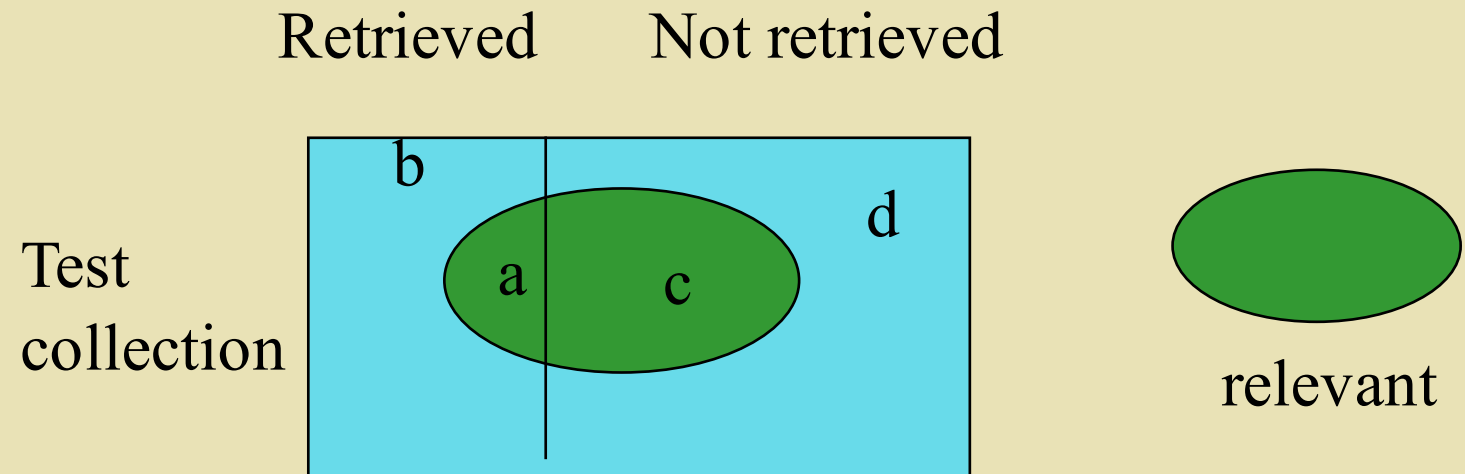
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Information Retrieval Techniques

- ◆ Based on search methods:
 - Text pattern search
 - Inverted files
 - Signature files
- ◆ Based on conceptual models:
 - Exact match
 - Text pattern, Boolean search
 - Inexact match
 - Probabilistic, vector space, inference networks

Recall and Precision



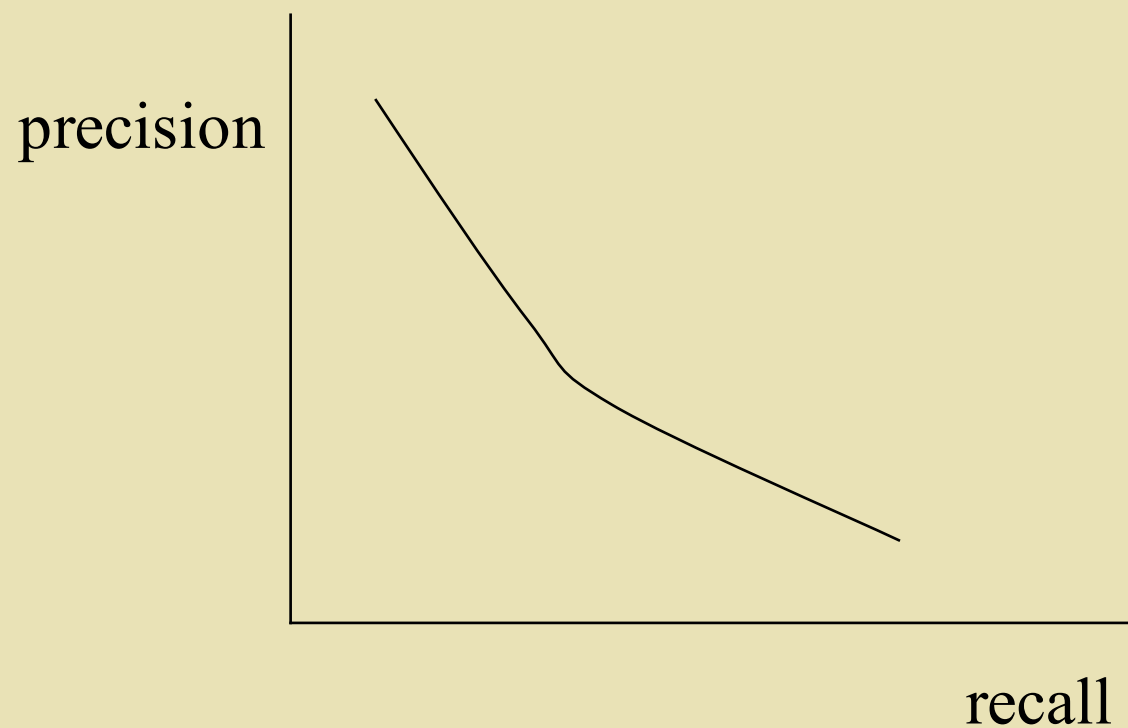
$$\text{Recall} = a / a + c$$

$$\text{Precision} = a / a + b$$

$$\text{Cut-off} = a + b / a + b + c + d$$

$$\text{Fall-out} = b / b + d$$

Recall precision graph





Inverted file

- ◆ An inverted file is a sorted file of keywords from a document collection, in which each keyword is associated with links to the documents containing that keyword
- ◆ Thus an inverted file provides an index to the documents based on the keywords selection



Inverted file example

	Term1	term2	term3	term4
Record1	1	1	0	1
Record2	0	1	1	1
Record3	1	0	1	1
Record4	0	0	1	1

	record1	record2	record3	record4
term1	1	0	1	0
term2	1	1	0	0
term3	0	1	1	1
term4	1	1	1	1



Signature file

- ◆ Signature file is based on the idea of inexact filtering, which quickly discards irrelevant documents for a given query.
- ◆ Hashed coded bit patterns
- ◆ Signature files requires 10~15% space overhead in contrast to 50~300% of inverted files
- ◆ Suitable for write once read many applications



Signature files

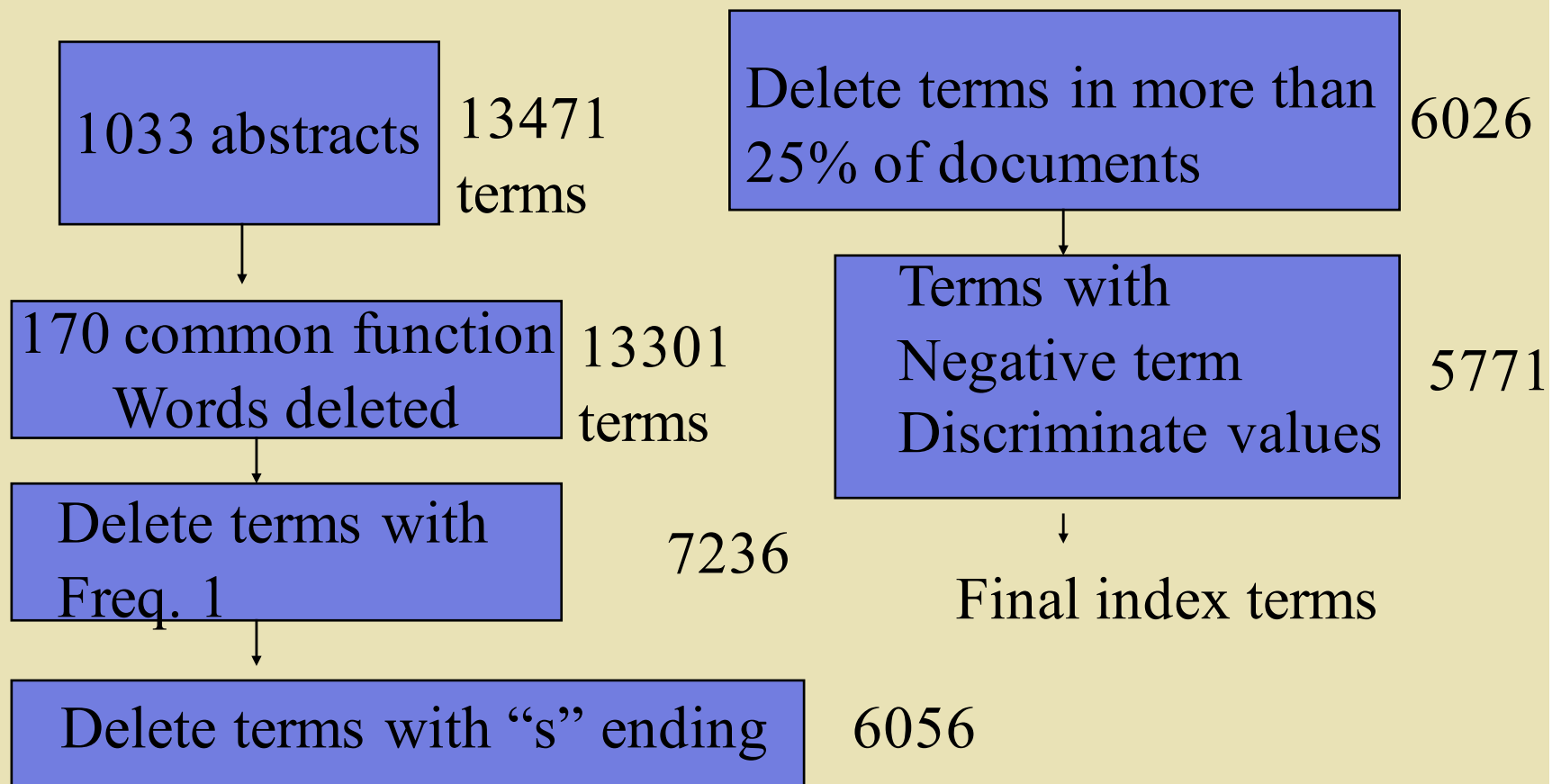
- ◆ Each documents is divided into logical blocks, each contains D distinct words
- ◆ Each word yields a word signature which is a bit pattern of size F with m bits set to 1
- ◆ The word signatures of each block is or-ed together to form the block signature
- ◆ Block signature concatenated to form a document signature



Simple techniques for text

- ◆ Stop list: article, prepositions, pronoun, conjunctions
- ◆ Stemming: suffix, prefix
- ◆ Use of Thesaurus: replacing or adding thesaurus categories as index terms

Example of selecting index terms





Retrieval models

- ◆ Boolean model:
 - AND,OR,NOT operations on index terms
 - Without ranking
- ◆ Vector space model:
 - $D_i=(a_{i1},a_{i2},\dots,a_{it})$
 - $Q_j=(q_{j1},q_{j2},\dots,q_{jt})$
 - All entries can be Boolean values or weightings
- ◆ Similarity between the two vectors can be computed as $D_i \cdot Q_j$ or cosine ...



Term Weighting Scheme

TF-IDF: Term Frequency-Inverse Document Frequency

- A well studied term weighting scheme in information retrieval
- Terms that occur in fewer documents are the better discriminators
- The importance of terms:
 - Proportional to the terms frequency
 - Inversely proportional to the number of documents to which the terms occur



TF-IDF Formula

$$w_i(d) = tf_i(d) \cdot \log\left(\frac{N}{df_i}\right)$$

$w_i(d)$: the weight of term i in document d

$tf_i(d)$: the frequency occurrence of term i in document d

N : the number of document in document
collection

df_i : the document frequency of term i (the number of
documents in the collection that contain term i)



Vector Similarity Functions

Measure the similarity between two vectors.

- Cosine
- Dice
- Jaccard
- Overlap measure
- Asymmetric measure

Cosine Similarity Measure

$$\text{sim}(d_i, d_j) = \frac{\sum_{k=1}^t [w_k(d_i) \cdot w_k(d_j)]}{\sqrt{\sum_{k=1}^t [w_k(d_i)]^2 \cdot \sum_{k=1}^t [w_k(d_j)]^2}}$$



Dice Similarity Measure

$$\text{sim}(d_i, d_j) =$$

$$\frac{2 \sum_{k=1}^t [w_k(d_i) \cdot w_k(d_j)]}{\sum_{k=1}^t w_k(d_i) + \sum_{k=1}^t w_k(d_j)}$$



Jaccard Similarity Measure

$$\text{sim}(d_i, d_j) =$$

$$\frac{\sum_{k=1}^t [w_k(d_i) \cdot w_k(d_j)]}{\sum_{k=1}^t w_k(d_i) + \sum_{k=1}^t w_k(d_j) - \sum_{k=1}^t [w_k(d_i) \cdot w_k(d_j)]}$$



Overlap Measure

$$\text{sim}(d_i, d_j) = \frac{\sum_{k=1}^t [w_k(d_i) \cdot w_k(d_j)]}{\min \left\{ \sum_{k=1}^t w_k(d_i), \sum_{k=1}^t w_k(d_j) \right\}}$$



Asymmetric Measure

$$\text{sim}(d_i, d_j) = \frac{\min_{k=1}^t \{w_k(d_i), w_k(d_j)\}}{\sum_{k=1}^t w_k(d_i)}$$



Relevance feedback

- ◆ $Q_{opt} = k \{ 1/|R| \sum_{D_i \in R} |D_i| - 1/(|N|) \sum_{D_j \in N} |D_j| \}$
- ◆ R is the set of relevant document; N is non-relevant one
- ◆ Reformulate the query to approximate the optimal query
- ◆ $Q^{(t+1)} = Q^{(t)} + 1/|R| \sum_{D_i \in R} d_i - 1/|N| \sum_{D_j \in N} D_j$



PageRank algorithm

- ◆ Google search engines use the algorithm to return web pages from query
- ◆ Page rank weights **higher quality page** a higher score

$$PR(p) \text{ is } (1-d)/N + d \sum_i PR(in_i)/C(in_i)$$

N is total number of pages in the corpus,

d is damping factor,

$C(in_i)$: count of the total number of **out-links on page** in_i

d= 0.85



HITS algorithm

- ◆ $\text{HITS}(query)$:
- ◆ $pages \leftarrow \text{EXPAND-PAGES}(\text{relevant-pages}(query))$
- ◆ For each p in $pages$ do
- ◆ $P.AUTHORITY \leftarrow 1$
- ◆ $P.HUB \leftarrow 1$
- ◆ Repeat until converge do
- ◆ for each p in $pages$ do
- ◆ $p.AUTHORITY \leftarrow \sum_i \text{INLINK}_i(p).HUB$
- ◆ $p.HUB \leftarrow \sum_i \text{OUTLINK}_i(p).AUTHORITY$
- ◆ $\text{Normalize}(pages)$
- ◆ Return $pages$



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Information integration

- ◆ Convert target domains into a unified domain model
- ◆ Construct a query plan that maps a query in the domain model to sub-queries against relevant data sources
- ◆ Each data source, there is a wrapper that opens and extracts data from source

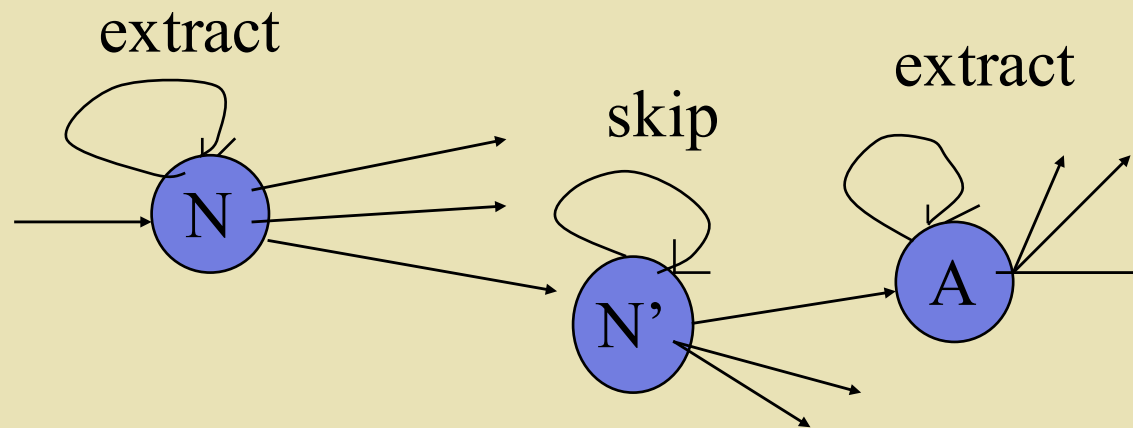


Web wrapper

- ◆ Web consists of semi-structured documents
- ◆ Web wrapper construction:
 - convert web into a standard database
 - require an information extractor
- ◆ SoftMealy[Hsu]: information extraction process as a finite state machine

Learning Extraction rules

- ◆ A contextual rule is like:
 - TRANSFER FROM state N to state N'
 - IF Left context = capitalizing string
 - Right context = HTML tag “

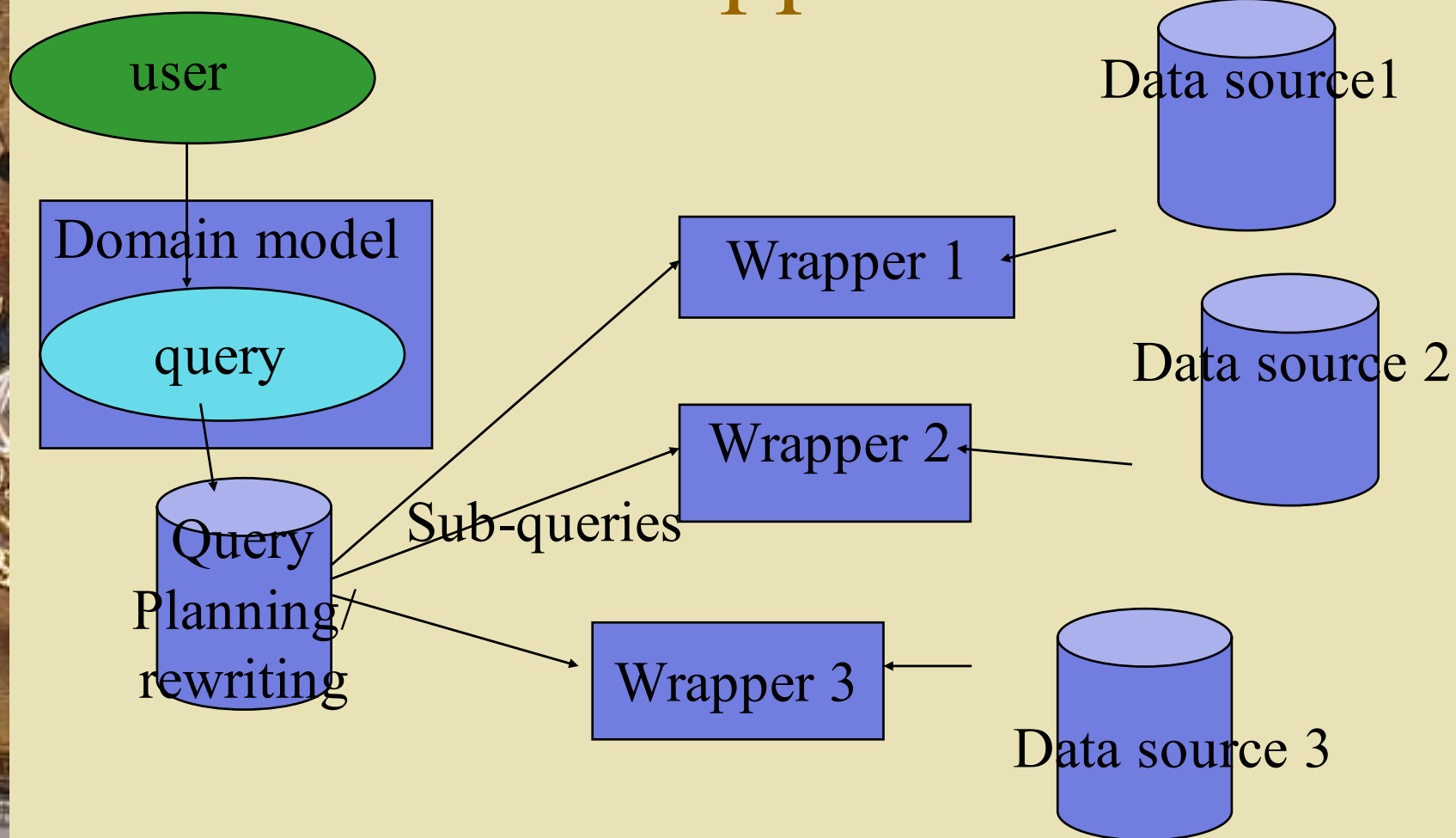




Data extractor

- ◆ Labeling component: users provide examples
- ◆ Learning component: learning data extraction rule
- ◆ Extraction component: execute the rule

Information integration with wrappers





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Information Filtering

- *Information explosion on the Internet*
- **Problem** : *choosing relevant information*
- **Characteristics of Information Content:**
 - *Documents semi-structured or unstructured (unlike database systems)*
 - *Lack of a standard way to describe contents of a document (unlike library systems)*

Information Filtering Technologies

- Collaborative Filtering
- Content-based Filtering
- Knowledge-based Filtering





Collaborative Filtering

- Makes a recommendation based on the correlation among users.
- Predicts a user's rating on an unseen item based on the rating of other users with similar interests.

Uses Pearson correlation coefficient to measure the correlation between two users.

Predicts the user's rating by averaging the product of each user rating on the unseen item and the correlation between users.



Collaborative Filtering

- ◆ Using other peers to recommend items to potential users
- ◆ Restaurant recommendation problem
 - N restaurants with N_r different attributes such as types, price, serve-wine, decoration, et
 - Ranking on N restaurants from K users
 - User profiles are described by N_u attributes such as sex, age, income-level, profession, etc.

How can be use the data to recommend a user
The ranking of a suitable restaurant?



Pearson Correlation Coefficient

$$r(x, y) = \frac{\sum_{k=1}^t (R_{x,k} - R_x)(R_{y,k} - R_y)}{\sqrt{\sum_{k=1}^t (R_{x,k} - R_x)^2 \cdot \sum_{k=1}^t (R_{y,k} - R_y)^2}}$$

$r(x, y)$: the Pearson correlation between users x and y

$R_{x,k}$: the rating of user x on item k

R_x : the average rating by user x

User's Rating Prediction

$$R_{x,k} = R_x + \frac{\sum_y (R_{y,k} - R_y) r(x, y)}{\sum_y |r(x, y)|}$$

$R_{x,k}$: the rating prediction for user x on unseen item k





Problem of collaborative filtering

- ◆ Coldstart problem: lack of user rankings



Firefly www.firefly.com

- ◆ Recommends music and films to users.
- ◆ Asks each user to rate a number of artists.
- ◆ Compare those ratings with ratings of other users.
- ◆ The underlying idea is like how we solicit recommendations from friends.
- ◆ Can be computationally costly for serving many users



Content-based Approach

- ◆ The weight of a word in a category is calculated based on
 - How many times does the word appear in a document? (The more often the higher the weight.)
 - How often does the word occur in other documents? (The more often the lower the weight)



Knowledge-based Approaches

- ◆ Use ontology and domain-specific knowledge to represent relationships between words and concepts.

IF the document contains “suicide bombing”
THEN the document is likely to be related to
“terrorism”.

IF the user likes “basketball” but not “College
basketball”

THEN the user is likely to like “NBA”



Content-based vs Knowledge-based Approaches

- ◆ Content-based Approach
 - Uses the content of a document to represent a user's interest.
 - Typically uses weighted keywords.
- ◆ Knowledge-based Approach
 - Uses knowledge about a problem domain (also called ontology) to infer a user's interest.
 - Typically uses rules.



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Personalized Information service and delivery

- ◆ Delegation of users
- ◆ Representation of user profile
 - Vector representation
 - Short term vs long term interest
 - Other background information
 - Collaborative filtering
- ◆ Mode of delivery
 - Phone
 - Wireless
 - Email
 - Web update (push or pull)



How to recommend?

- ◆ Principle: Similar user has similar taste?
- ◆ Find similarity between users based on user profile
- ◆ Find similarity of restaurants
- ◆ Transfer of ranking of users with similar profile to the ranking of similar restaurants

Syskill & Webert

- ◆ <http://www.ics.uci.edu/~pazzani/RTF/AAAI.html>





WebWatcher [Mitchell]

- ◆ A learning apprentice for WWW
 - Provide interactive advice
 - Automatically acquire knowledge for selecting hyperlink given current web page viewed by the user and user's information goal
- ◆ Recommend next hyperlink at current page

What should be learned?

◆ LinkUtility:

Page*Goal*User*Link [0,1]

- Page: current page
- Goal: information seek by the user
- User: identity of user
- Link: one of the hyperlinks found on Page





Feature Selection for WebWatcher

- ◆ The Goal, Page, Links are represented as vectors in terms of 530 features

200 words underlined	200 words sentence	100 words heading	~30 words User goal
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Homework 5 (Due June 12)

- ◆ Read a paper on recommender systems
- ◆ Write down the authors and their affiliations
- ◆ Write down the source of the paper
- ◆ Describe what items they are recommending and discuss what methods they use
- ◆ Make your critical comments on the paper
- ◆ You should not use more than one page